

Mohammed Ali Safwan Khan

BEng (Hons) Electrical & Electronic Engineering · Aston University

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PROFILE

Third-year BEng (Hons) Electrical and Electronic Engineering student at Aston University, with hands-on experience in embedded systems, control theory, PCB design, and digital electronics. Student member of the Institution of Engineering and Technology (IET). Seeking a placement opportunity to apply technical and practical skills within a high-performance engineering team.

EDUCATION

BEng (Hons) Electrical and Electronic Engineering

2023 – 2026

Aston University · Birmingham

- IET-accredited integrated degree with required placement year
- Modules include: Computational Problem Solving, Digital Design (VHDL), Control Systems, Electric Drive Train Control, Embedded Systems
- Skills gained: MATLAB/Simulink, Proteus, Python, C/C++, FreeRTOS, LabVIEW, VHDL

BTEC Level 3 National Extended Diploma in Engineering (D* D* D)

2020 – 2023

Aston University Engineering Academy · Birmingham

- Key units: Computer-Aided Design, Specialist Engineering Project, Electronic Devices and Circuits

GCSEs

2015 – 2020

Holly Lodge High School

- 6 GCSEs (Grades 3–5) including English Language (4) and Mathematics (5)

PROJECTS

Traction Control System for an RC Vehicle

2025 – 2026

Final Year Dissertation · Aston University

- Designed and built a full TCS from the ground up: custom manufactured PCB, dual-core ESP32 firmware (FreeRTOS), and MATLAB/Simulink control model
- Achieved 50–63% wheel slip reduction at low/mid throttle on wood surface, exceeding the 25% SMART objective

2-Player NeoPixel Snake

2023 -- 2024

University Team Project · Aston University

- Team project building a two-player Snake game on a custom NeoPixel display: two 16×8 panels connected in series, driven by an Arduino Mega
- Role: Electronic Procurement Lead — responsible for selecting, sourcing, and testing all components, including LED panels, Arduino Mega, power supply, and joysticks

Audio Amplifier

2023

Individual Project · Aston University

- End-to-end project: amplifier circuit designed from first principles (gain, biasing, frequency response), simulated in Proteus, then physically manufactured and tested

Pencil Case Speaker

2022

Specialist Project · Aston University Engineering Academy

- Designed an enclosure in Fusion 360 and fabricated to house both stationery and a working Kitronics speaker system

TECHNICAL SKILLS

Firmware

C / C++ · C (AVR) · FreeRTOS · ESP-IDF · Arduino

Control Systems

PID Design · MATLAB · Simulink · SimPowerSystems

Hardware Design

PCB Layout · Proteus · PCB Fabrication · Schematic Capture

Digital Design

VHDL · Logic Design · Digital Circuits

Instrumentation

LabVIEW · Oscilloscope · Logic Analyser · IMU

Programming

Python · Object-Oriented Design

Data Analysis

Matplotlib · CSV Telemetry · NumPy

Office

Microsoft Word · Excel · PowerPoint · Publisher

INTERESTS & MEMBERSHIPS

- IET Student Member · Aston Electronics Society · Aston Islamic Society · Self-study: control systems theory · Full UK driver's licence